



Maheshchandra Hegde

Founder & Technology Leader at Technoayan Digital Transformation Services Pvt. Ltd. | Building AI-driven 3D asset platforms and 3d computing solutions
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Executive Introduction: Innovative technologist and product strategist with **18+ years of experience** architecting end-to-end digital products, interactive 3D experiences, and domain-specific SaaS platforms. Co-Founder and Technology Leader at **Technoayan Digital Transformation Services Pvt. Ltd.**, driving product architecture, cloud scalability, and creative technical vision across modern web/mobile application stacks, interactive 3D/PBR pipelines, real-time spatial computing, and design-to-code automation across both high-velocity startups and mission-critical enterprise systems.

PROJECT EXPERIENCE & TECHNICAL TRACK RECORD

1. Projects Executed as Founder at Technoayan Digital Transformation Services Pvt. Ltd.

Voxelforge AI (srushtilabs.com/voxelforge/)

Developed Voxelforge AI, a generative AI platform creating modular 3D assets from user prompt inputs. Generates game-ready, low-poly modular 3D models and asset bundles optimized for immediate assembly in Unreal Engine, Unity, and real-time WebGL engines, cutting prototyping turnaround from days to minutes.

Ayam3d (ayam3d.in)

Extended the Voxelforge concept into Ayam3d ('Ayam' meaning Dimension), generating parametric-based 3D models from natural language prompts in seconds using custom trained AI models. Received early investor interest; actively developing and seeking strategic partners to scale the AI pipeline.

2. Projects Executed at Cyient Limited / PHILIPS Healthcare

Philips Intellispace Critical Care & Anesthesia (ICCA)

Served as Senior Technology Leader (React / NodeJS) delivering hospital ICU software products. Challenging 24/7 mission-critical healthcare environment passing stringent quality and safety benchmarks set by Philips and hospital networks. Security of the application was critical: strictly enforced Multi-Factor Authentication (MFA), Cross-Site Scripting (XSS) prevention, role-based access control (RBAC), and patient data protection. Delivered end-to-end best practices, modular architecture, and production services.

3. Projects Executed at Ness Digital Engineering

Entertainment Management Application

As Senior Analyst / Tech Lead, led a development squad to build an Entertainment Management Application using Angular, TypeScript, Node.js, and AWS. Architected the complete single-page application structure, enforced accessibility standards (WCAG) and code quality linters, and took technical leadership in deploying and delivering the application to the client on schedule.

4. Projects Executed at Moonraft Innovation Labs (Unit of UST Global)

Enterprise Cross-Framework UI Design System

Built, architected, and engineered a centralized design system using LitElement Web Components, React, AngularJS, Node.js, and npm packaging. Stored all standardized UI components in a central repository, eliminating reinventing the wheel across client projects and guaranteeing brand coherence from a single codebase.

Luxury Hotel Group Mobile Application

Led mobile engineering team to develop a cross-platform mobile booking application for a premier Indian hotel group (TLC Group of Hotels) built using Ionic and Angular, delivering high performance and polished guest experiences.

5. Projects Executed at UST Global / CISCO

Cisco Stadium Vision Director Platform Modernization

Senior Systems Analyst and Associate Project Manager for 4+ years on Cisco Stadium Vision Director—an enterprise digital media distribution platform deployed in major international stadiums. Took technical leadership in systematically migrating the legacy Flash and Flex platform to modern Angular and React modules step by step. Comprised numerous sub-applications with active global clients; successfully executed the migration without disrupting live stadium operations. Recognized with multiple UST Global Excellence Awards.

6. Projects Executed at ThoughtFocus Technologies

Aftermarket Parts Explorer Application

UI Architect and Tech Lead for an enterprise aftermarket parts procurement platform supporting millions of parts from diverse manufacturers. Built an interactive 3D viewer allowing buyers to inspect 3D models and dimensions of thousands of components, integrated with cart and payment gateways (Adobe Flex, Mate, ActionScript, HTML).

Enterprise UI Solutions & Bootstrapping

Established early user prototyping pipelines. Guided team to bootstrap high-speed enterprise search applications and an operational logistics management application for a large milk/dairy corporation (HTML, CSS, JavaScript, TypeScript, Flex).

7. Facility Dashboard Development at CISCO & Vodafone (Independent Consultant)

Unified Building Management Systems (BMS) Dashboard

Designed and developed an interactive facility dashboard at Cisco Systems campus as a full-time consultant (2009–2010). Developed a unified web interface connecting to and controlling diverse Building Management Systems: Air Handling Units (AHUs), Variable Refrigerant Volume (VRVs), industrial chillers, and environmental sensor telemetry. Awarded formal Certificate of Appreciation from Cisco in 2010. Also delivered data visualization dashboards for Vodafone operations.

8. CAE Simulation Technologies (Canadian Aeronautical Engineering)

Flight Simulator Visual Database Development

Worked as a Visual Database Developer (Sep 2007 – Apr 2008) at CAE, the global leader in civil and military aircraft flight simulators. Purely visual and structural design work where visual, topographical, and structural details were modeled and fed into databases used by real-time flight simulators. Created high-fidelity 3D synthetic environments (runways, airports, terrain) using 3D modeling packages, Adobe Photoshop for procedural texture generation, and strict Level of Detail (LOD) hierarchies under locked 60 FPS simulator budgets.

9. Other Specialized Engineering Projects (Technoyana Digital)

Apart from traditional software development, led and delivered diverse multi-disciplinary projects: architectural 3D visualization and animated walkthroughs; customized CNC machine development and automated CAD/CAM toolpath pipelines; built consumer mobile application *SellAny*; and led development of an automated parking payment mobile application with real-time slot occupancy tracking.

MAJOR ACHIEVEMENTS & MILESTONES

1. Zero-Downtime Global Stadium Platform Migration

Modernized Cisco Stadium Vision Director deployed at international sporting arenas across a 4-year tenure. Led phased migration from Adobe Flash/Flex to Angular/React without a single minute of venue downtime. Awarded 3 UST Global Excellence Awards (2015, 2016, 2018).

2. Healthcare-Grade ICU Software Delivery (Philips)

Delivered web-tier clinical software for Philips ICCA in 24/7 ICU suites. Passed stringent hospital quality, safety, and zero-leak reliability benchmarks for bedside patient monitoring. Enforced MFA, RBAC, and strict medical data protection.

3. Generative 3D Asset & Spatial AI (SrushtiLabs)

Founded and launched Voxelforge AI (srushtilabs.com/voxelforge/) creating modular, game-ready 3D bundles for Unreal Engine, Unity, and WebGL. Developed Ayam3d (ayam3d.in) for parametric 3D model generation, earning early investor interest.

4. Enterprise UI Design System & Cisco BMS Telemetry

Architected centralized Web Components design system (LitElement, React, Angular) at Moonraft / UST, eliminating duplicated UI effort across teams. Designed unified Cisco facility telemetry dashboard; awarded formal Cisco Certificate of Appreciation (2010).

CHALLENGES FACED & RESOLUTIONS IMPLEMENTED

Legacy Codebase Modernization (Cisco)

Challenge: Adobe Flash/Flex deprecation required migrating complex global stadium platform without venue downtime.

Resolution: Built hybrid micro-frontend event bridge allowing legacy and modern React/Angular modules to coexist and migrate incrementally.

Continuous Bedside Reliability (Philips ICU)

Challenge: 24/7 multi-day operation without browser memory leaks or UI freezes under dense patient telemetry.

Resolution: Enforced rigorous memory profiling, deterministic component cleanup on unmount, and automated regression test gates.

Fragmented UI Development (Moonraft)

Challenge: Multiple squads independently building identical UI widgets across React and Angular, causing brand drift.

Resolution: Architected centralized LitElement design system packaged via npm, standardizing components across all teams.

Generative 3D Mesh Topology (SrushtiLabs)

Challenge: Raw AI 3D meshes produced chaotic vertex topology and high polycounts unsuited for real-time rendering.

Resolution: Built automated retopology, polygon reduction, UV unwrapping, and texture baking pipeline for lightweight game-ready assets.

LESSONS LEARNED & BEST PRACTICES

Incremental Modernization Over Rewrites

Complete rewrites carry high operational risk and delayed value. Strangler-fig micro-frontends and backward-compatible APIs enable continuous delivery with zero downtime.

Security & Compliance from Day Zero

Retrofitting security late causes friction and vulnerabilities. Embed MFA, RBAC, automated vulnerability scanning, and audit logging into the CI/CD pipeline from inception.

Design Systems as Strategic Products

Treat shared UI libraries as internal products with semantic versioning and framework-agnostic standards to compound ROI and brand consistency across squads.

Interactive Prototyping Resolves Ambiguity

Early functional mockups and clickable UI prototypes validate operational workflows with users and leadership, preventing costly changes post-deployment.

SUGGESTIONS FOR FUTURE PROCESS IMPROVEMENTS

Automated IT Governance & Self-Service IDPs

Establish standardized developer platforms and Infrastructure-as-Code (IaC) templates, cutting environment provisioning time from weeks to hours with built-in security baselines.

AI-Assisted Engineering Toolchains

Integrate generative AI tooling for automated test generation, synthetic data simulation, and design-to-code pipelines, boosting engineering throughput by 30–40%.

Unified Enterprise Telemetry & APM

Combine application monitoring, network metrics, and facility sensor data (BMS) into unified dashboards to transition IT operations from reactive fix to predictive observability.

Structured Upskilling & Certification

Institute regular cross-functional architecture reviews and sponsored certification tracks (Cloud, Security, ITIL, Agile) to ensure continuous organizational agility.

SUMMARY

Synthesis of Design & Engineering Mastery

As a graduate in Human Interface Design (M.S. in Computing, UK) and holding an engineering degree in Electronics and Communication (B.E.), I possess a balanced, end-to-end mastery of both the design and software development aspects of digital product creation with vast experience across the entire product lifecycle.

Proven Versatility Across Startup Velocity & Enterprise Scale

My 18+ years of experience encompass handling both small and large teams and projects at various stages of product maturity—from fast-paced startup incubation (Technoyana, SrushtiLabs, Twitan) to large-scale, safety-critical enterprise systems (Philips ICU Healthcare, Cisco Systems, UST Global, Ness).

Pledge for Self-Training & Professional Certification

As per the requirements of the **Senior Manager – IT** role, I pledge to proactively upskill myself for the needs of the team, role, and organizational responsibilities.

I agree to undertake and complete any required professional and enterprise certifications (e.g., Cloud Architecture, ITIL, Cybersecurity/CISSP, or Project Governance) as per project and team needs before and upon joining.

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